

Global Coal Power & Emissions R Project

Logan Rosell

```
library(ggplot2)
library(tidyverse)
```

Warning: package 'tidyr' was built under R version 4.2.3

Warning: package 'readr' was built under R version 4.2.3

Warning: package 'dplyr' was built under R version 4.2.3

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v lubridate  1.9.4      v tibble     3.3.0
v purrr      1.1.0      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(skimr)
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
library(gtsummary)
library(gt)
library(lawstat)
```

Attaching package: 'lawstat'

The following object is masked from 'package:car':

levene.test

```
library(sf)
```

Warning: package 'sf' was built under R version 4.2.3

Linking to GEOS 3.11.0, GDAL 3.5.3, PROJ 9.1.0; sf_use_s2() is TRUE

```
library(rnaturalearth)
library(extrafont)
```

Registering fonts with R

```
#font_import()
#loadfonts(device = "pdf")

theme_set(theme_classic())
theme_update(text = element_text(family = "Times New Roman"))

setwd("~/Desktop/WU R Class/R Project Files")
```

Data Source: <https://ember-energy.org/data/yearly-electricity-data/>

Question: Does coal power's CO2 emission intensity vary across global regions and time between 2000 and 2024?

Cleaning Data

```
global_power_data <- read_csv("yearly_full_release_long_format.csv")
```

Rows: 357076 Columns: 18

-- Column specification -----
Delimiter: ","

chr (9): Area, ISO 3 code, Area type, Continent, Ember region, Category, Sub...

dbl (9): Year, EU, OECD, G20, G7, ASEAN, Value, YoY absolute change, YoY % c...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
global_coal <- global_power_data %>%  
  filter(Variable == 'Coal') %>%  
  filter(Category != "Capacity") %>%  
  dplyr::select(-c(`YoY absolute change`, `YoY % change`, Category, Subcategory, Variable)) %>%  
  pivot_wider(names_from = Unit, values_from = Value) %>%  
  mutate(mtCO2_per_TWh = mtCO2/TWh,  
         mtCO2_per_TWh = ifelse(is.na(mtCO2_per_TWh), 0, mtCO2_per_TWh))
```

Overview of Data

```
skim(global_power_data)
```

Table 1: Data summary

Name	global_power_data
Number of rows	357076
Number of columns	18
Column type frequency:	
character	9
numeric	9
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Area	0	1.00	2	44	0	228	0
ISO 3 code	21450	0.94	3	3	0	215	0
Area type	0	1.00	6	18	0	2	0
Continent	21450	0.94	4	13	0	6	0
Ember region	21450	0.94	4	27	0	7	0
Category	0	1.00	8	22	0	5	0
Subcategory	0	1.00	4	19	0	7	0
Variable	0	1.00	3	37	0	21	0
Unit	0	1.00	1	8	0	6	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Year	0	1.00	2011.69	7.02	2000.00	2006	2012.00	2018.00	2024.00	
EU	21450	0.94	0.13	0.34	0.00	0	0.00	0.00	1.00	
OECD	21450	0.94	0.18	0.39	0.00	0	0.00	0.00	1.00	
G20	21450	0.94	0.09	0.28	0.00	0	0.00	0.00	1.00	
G7	21450	0.94	0.03	0.18	0.00	0	0.00	0.00	1.00	
ASEAN	21450	0.94	0.05	0.21	0.00	0	0.00	0.00	1.00	
Value	19705	0.94	70.22	596.96	-89.85	0	0.25	8.65	30928.54	
YoY absolute change	111179	0.69	2.02	27.90	-	0	0.00	0.06	1530.71	
YoY % change	192883	0.46	11.14	194.22	-	0	0.81	9.09	24000.00	

```
summary(global_power_data)
```

Area	ISO 3 code	Year	Area type
Length:357076	Length:357076	Min. :2000	Length:357076
Class :character	Class :character	1st Qu.:2006	Class :character
Mode :character	Mode :character	Median :2012	Mode :character
		Mean :2012	
		3rd Qu.:2018	
		Max. :2024	

Continent	Ember region	EU	OECD
-----------	--------------	----	------

Length:357076	Length:357076	Min. :0.000	Min. :0.000
Class :character	Class :character	1st Qu.:0.000	1st Qu.:0.000
Mode :character	Mode :character	Median :0.000	Median :0.000
		Mean :0.132	Mean :0.184
		3rd Qu.:0.000	3rd Qu.:0.000
		Max. :1.000	Max. :1.000
		NA's :21450	NA's :21450

G20	G7	ASEAN	Category
Min. :0.000	Min. :0.000	Min. :0.000	Length:357076
1st Qu.:0.000	1st Qu.:0.000	1st Qu.:0.000	Class :character
Median :0.000	Median :0.000	Median :0.000	Mode :character
Mean :0.089	Mean :0.034	Mean :0.046	
3rd Qu.:0.000	3rd Qu.:0.000	3rd Qu.:0.000	
Max. :1.000	Max. :1.000	Max. :1.000	
NA's :21450	NA's :21450	NA's :21450	

Subcategory	Variable	Unit	Value
Length:357076	Length:357076	Length:357076	Min. : -89.85
Class :character	Class :character	Class :character	1st Qu.: 0.00
Mode :character	Mode :character	Mode :character	Median : 0.25
			Mean : 70.22
			3rd Qu.: 8.65
			Max. :30928.54
			NA's :19705

YoY absolute change	YoY % change
Min. :-500.00	Min. :-51275.00
1st Qu.: 0.00	1st Qu.: 0.00
Median : 0.00	Median : 0.81
Mean : 2.02	Mean : 11.14
3rd Qu.: 0.06	3rd Qu.: 9.09
Max. :1530.71	Max. : 24000.00
NA's :111179	NA's :192883

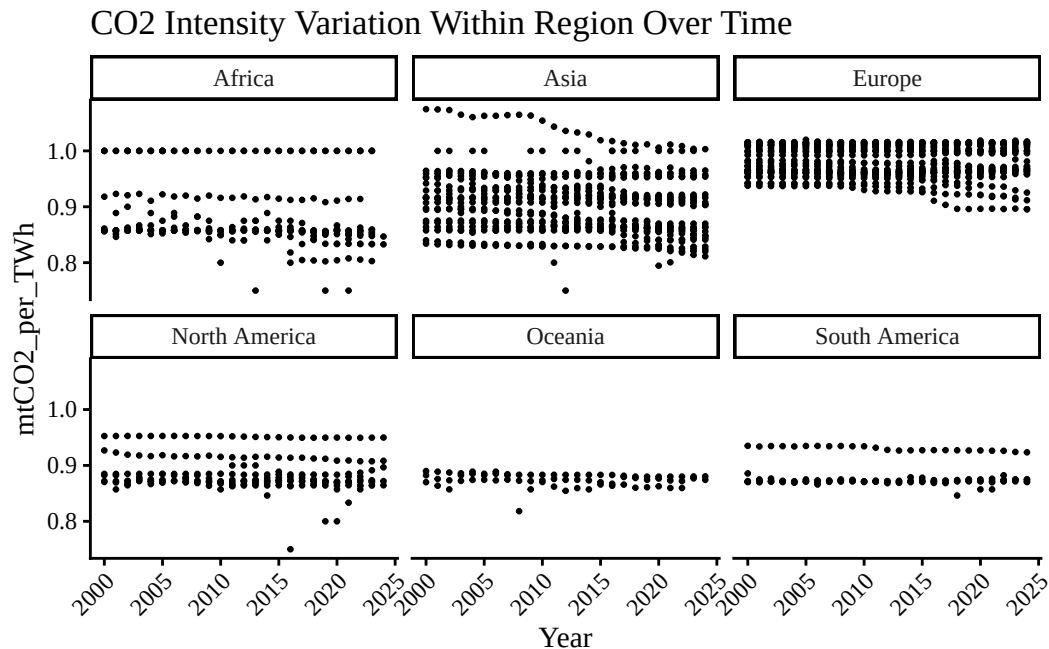
```
countries <- global_power_data %>% filter(`Area type`=="Country or economy") %>% dplyr::select(
  unique(countries)
```

```
# A tibble: 215 x 1
  Area
  <chr>
1 Afghanistan
2 Albania
3 Algeria
4 American Samoa
```

```
5 Angola
6 Antigua and Barbuda
7 Argentina
8 Armenia
9 Aruba
10 Australia
# i 205 more rows
```

Region Analysis

```
global_coal %>%
  filter(TWh != 0) %>%
  filter(`Area type` == "Country or economy") %>%
  ggplot(aes(x=Year, y=mtCO2_per_TWh)) +
  geom_point(size=0.5) +
  facet_wrap(~Continent) +
  theme(
    axis.text.x = element_text(
      angle = 45,
      vjust = 1,
      hjust = 1
    )
  ) +
  labs(title="CO2 Intensity Variation Within Region Over Time")
```



```
ggsave("CO2 Intensity Variation Within Region Over Time.png", width = 6, height=4, units = "in")
```

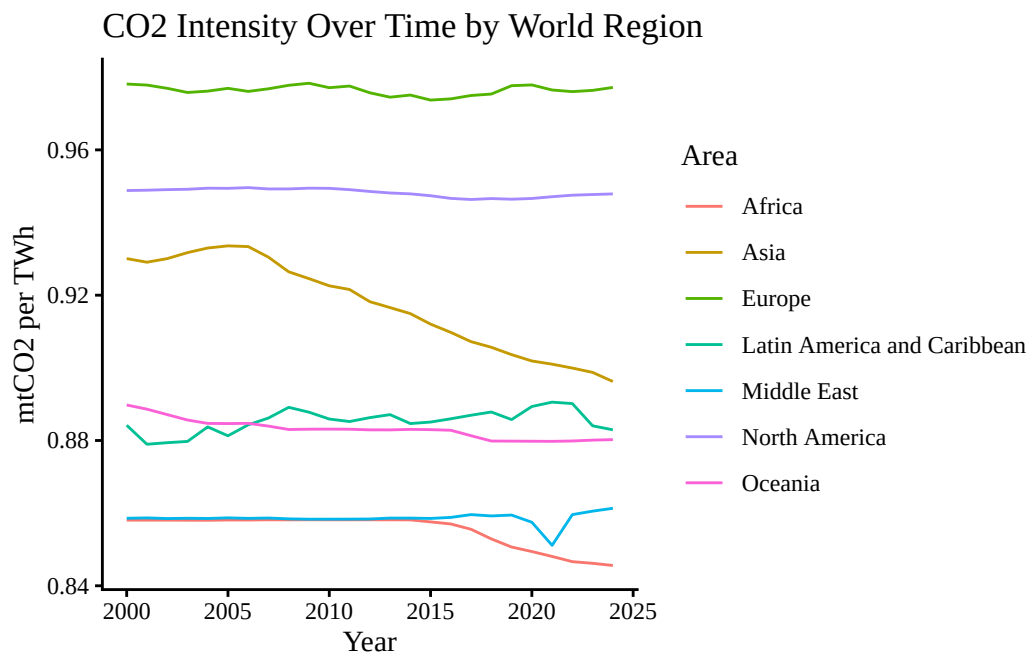
```
region_data <- global_coal %>%
  filter(`Area type` == 'Region') %>%
  filter(Area %in% c("Africa",
                    "Asia",
                    "Europe",
                    "Latin America and Caribbean",
                    "Middle East",
                    "North America",
                    "Oceania")) %>%
  dplyr::select(-c(`ISO 3 code`, Continent, `Ember region`, EU, OECD, G20, G7, ASEAN)) %>%
  mutate(percent_100 = `%` / 100)
```

```
table_1 <- region_data %>%
  tbl_summary(by = Area,
             include = c(`%`, TWh, mtCO2, mtCO2_per_TWh),
             statistic = all_continuous() ~ "{mean} ({median})") %>%
  as_gt()
gtsave(data = table_1, filename = "summary_stats_table_by_region.png")

table_2 <- region_data %>%
```

```
tbl_summary(by = Year,
            include = c(`%`, TWh, mtCO2, mtCO2_per_TWh),
            statistic = all_continuous() ~ "{mean} ({median})") %>%
  as_gt()
gtsave(data = table_2, filename = "summary_stats_table_by_year.png")
```

```
ggplot(region_data, aes(x=Year, y=mtCO2_per_TWh, color=Area)) +
  geom_line() +
  labs(title="CO2 Intensity Over Time by World Region",
       y= "mtCO2 per TWh")
```



```
ggsave("CO2 Intensity Over Time by World Region.png", width = 6, height=4, units = "in", dpi
```

Summarizing Region Data

```
skim(region_data)
```

Table 4: Data summary

Name	region_data
Number of rows	175
Number of columns	8
Column type frequency:	
character	2
numeric	6
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Area	0	1	4	27	0	7	0
Area type	0	1	6	6	0	1	0

Variable type: numeric

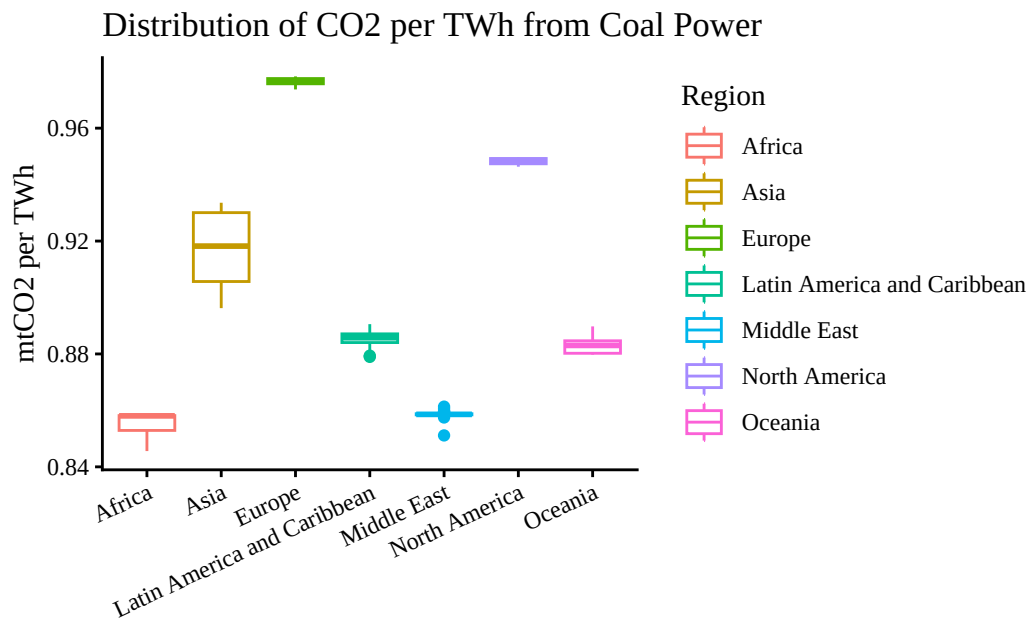
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Year	0	1	2012.00	7.23	2000.00	2006.00	2012.00	2018.00	2024.00	
%	0	1	30.52	21.15	0.67	6.42	27.49	48.25	68.95	
TWh	0	1	1224.50	1965.53	10.24	96.88	253.60	1292.51	8738.09	
mtCO2	0	1	1132.27	1788.97	8.82	85.97	217.63	1245.28	7831.36	
mtCO2_per_TWh	0	1	0.90	0.04	0.85	0.86	0.89	0.95	0.98	
percent_100	0	1	0.31	0.21	0.01	0.06	0.27	0.48	0.69	

```
summary(region_data)
```

Area	Year	Area type	%
Length:175	Min. :2000	Length:175	Min. : 0.67
Class :character	1st Qu.:2006	Class :character	1st Qu.: 6.42
Mode :character	Median :2012	Mode :character	Median :27.49
	Mean :2012		Mean :30.52
	3rd Qu.:2018		3rd Qu.:48.26
	Max. :2024		Max. :68.95
TWh	mtCO2	mtCO2_per_TWh	percent_100
Min. : 10.24	Min. : 8.82	Min. :0.8456	Min. :0.0067

1st Qu.:	96.89	1st Qu.:	85.97	1st Qu.:	0.8590	1st Qu.:	0.0642
Median :	253.60	Median :	217.63	Median :	0.8862	Median :	0.2749
Mean :	1224.50	Mean :	1132.27	Mean :	0.9034	Mean :	0.3052
3rd Qu.:	1292.51	3rd Qu.:	1245.29	3rd Qu.:	0.9472	3rd Qu.:	0.4825
Max. :	8738.09	Max. :	7831.36	Max. :	0.9783	Max. :	0.6895

```
ggplot(region_data, aes(x=Area, y=mtCO2_per_TWh, color=Area)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1)) +
  labs(x="",
       title="Distribution of CO2 per TWh from Coal Power",
       color = "Region",
       y= "mtCO2 per TWh")
```



```
ggsave("Distribution of CO2 per TWh from Coal Power.png", width = 7, height=4, units = "in",
```

Summary Stats by Region

```
region_data %>%
  group_by(Area) %>%
```

Area	N	Mean	Median	SD	Min	Max
Africa	25	0.855	0.858	0.005	0.846	0.858
Asia	25	0.917	0.918	0.013	0.896	0.934
Europe	25	0.976	0.976	0.001	0.974	0.978
Latin America and Caribbean	25	0.885	0.886	0.003	0.879	0.891
Middle East	25	0.859	0.859	0.002	0.851	0.861
North America	25	0.948	0.949	0.001	0.946	0.950
Oceania	25	0.883	0.883	0.003	0.880	0.890

```
summarize(
  N = n(),
  Mean = mean(mtCO2_per_TWh),
  Median = median(mtCO2_per_TWh),
  SD = sd(mtCO2_per_TWh),
  Min = min(mtCO2_per_TWh),
  Max = max(mtCO2_per_TWh)
) %>%
mutate(across(where(is.numeric), round, 3)) %>%
gt()
```

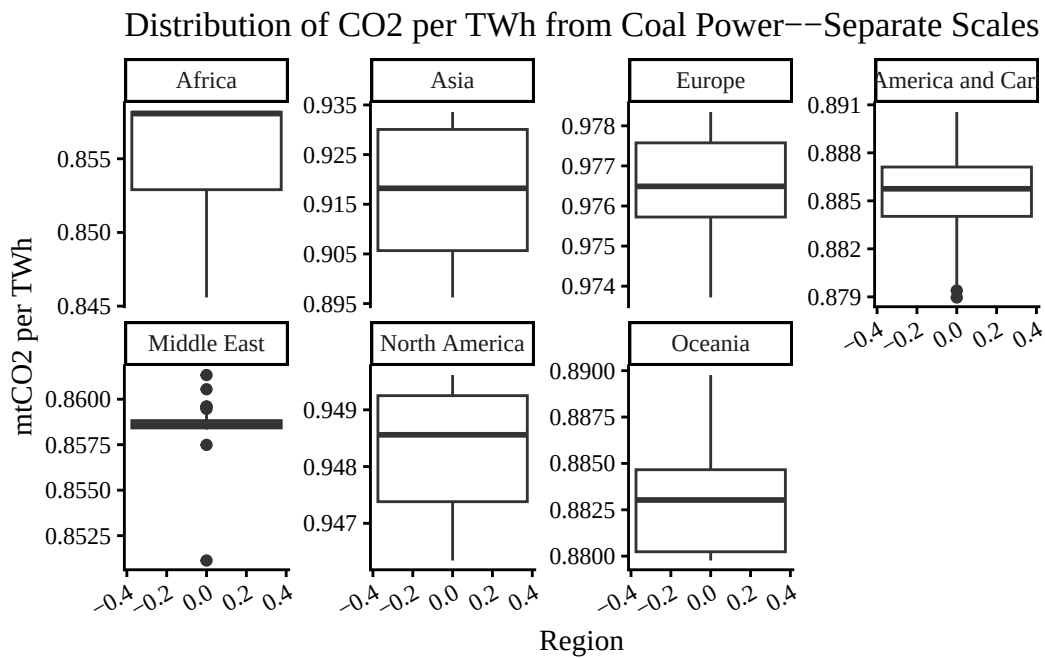
Warning: There was 1 warning in `mutate()`.
 i In argument: `across(where(is.numeric), round, 3)`.
 Caused by warning:
 ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
 Supply arguments directly to `.fns` through an anonymous function instead.

```
# Previously
across(a:b, mean, na.rm = TRUE)

# Now
across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
ggplot(region_data, aes(y=mtCO2_per_TWh)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 30, vjust = 1, hjust=1)) +
  labs(x="Region",
       title="Distribution of CO2 per TWh from Coal Power--Separate Scales",
       color = "Region",
       y= "mtCO2 per TWh") +
  facet_wrap(~ Area, nrow = 2, scales = "free_y")
```

Ignoring unknown labels:
 * colour : "Region"

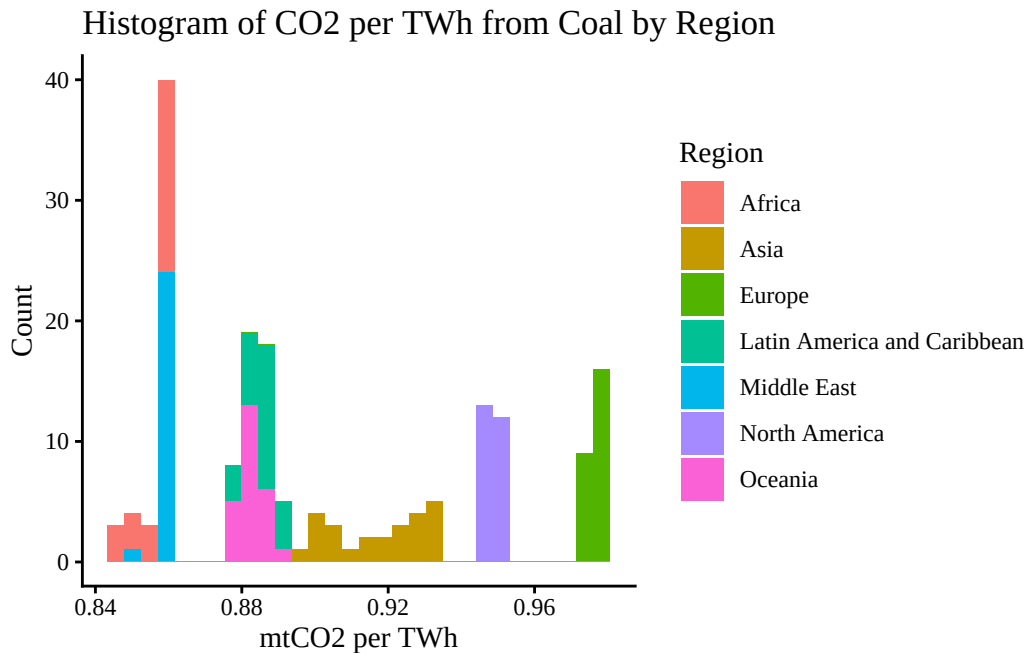


```
ggsave("Distribution of CO2 per TWh from Coal Power--Separate Scales.png", width = 7, height = 7)
```

Ignoring unknown labels:
 * colour : "Region"

```
ggplot(region_data, aes(x=mtCO2_per_TWh, fill=Area)) +
  geom_histogram() +
  labs(title="Histogram of CO2 per TWh from Coal by Region",
       fill="Region",
       y="Count",
       x="mtCO2 per TWh")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ggsave("Histogram of CO2 per TWh from Coal by Region.png", width = 7, height=4, units = "in"
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

Testing the difference in CO2 Intensity between regions

Null Hypothesis B: Coal power CO2 emission intensity does NOT vary between global regions.

Alternative Hypothesis B: Coal power CO2 emission intensity DOES vary between global regions.

```
shapiro.test(region_data$mtCO2_per_TWh)
```

Shapiro-Wilk normality test

```
data: region_data$mtCO2_per_TWh
W = 0.87914, p-value = 1.117e-10
```

A small p-value from our Shapiro-Wilks test tells us we should reject the null that our CO2_per_TWh data is normally distributed.

```
levene.test(region_data$mtCO2_per_TWh, group=region_data$Area)
```

Modified robust Brown-Forsythe Levene-type test based on the absolute deviations from the median

```
data: region_data$mtCO2_per_TWh
Test Statistic = 36.197, p-value < 2.2e-16
```

Small P-value for Brown-Forsythe means we reject the null of equal variance.

```
kruskal.test(mtCO2_per_TWh ~ Area, data=region_data)
```

Kruskal-Wallis rank sum test

```
data: mtCO2_per_TWh by Area
Kruskal-Wallis chi-squared = 167.33, df = 6, p-value < 2.2e-16
```

Because we don't have normally distributed data or equal variance I used the Kruskal-Wallis test (robust to both issues) and got a small p-value allowing us to reject the null that the distributions and medians are the same across global regions.

```
pairwise.wilcox.test(region_data$mtCO2_per_TWh, g = region_data$Area)
```

Pairwise comparisons using Wilcoxon rank sum exact test

```
data: region_data$mtCO2_per_TWh and region_data$Area
```

	Africa	Asia	Europe	Latin America and Caribbean
Asia	3.3e-13	-	-	-
Europe	3.3e-13	3.3e-13	-	-
Latin America and Caribbean	3.3e-13	3.3e-13	3.3e-13	-
Middle East	2.6e-09	3.3e-13	3.3e-13	3.3e-13
North America	3.3e-13	3.3e-13	3.3e-13	3.3e-13
Oceania	3.3e-13	3.3e-13	3.3e-13	0.0085

	Middle East	North America
Asia	-	-
Europe	-	-
Latin America and Caribbean	-	-
Middle East	-	-
North America	3.3e-13	-
Oceania	3.3e-13	3.3e-13

P value adjustment method: holm

All Pairs of regions are statically significantly different from each other.

Mapping the data (2024)

```
only_2024 <- global_coal %>%
  filter(Year == "2024") %>%
  filter(`Area type` == "Country or economy") %>%
  mutate(iso_3 = toupper(trimws(`ISO 3 code`))) %>%
  mutate(log_intensity = log((mtCO2_per_TWh * 100)))
```

```
world_map <- ne_countries(scale = "medium", returnclass = "sf")

map_data_2024 <- world_map %>%
  left_join(only_2024, by = c("iso_a3" = "iso_3"))
```

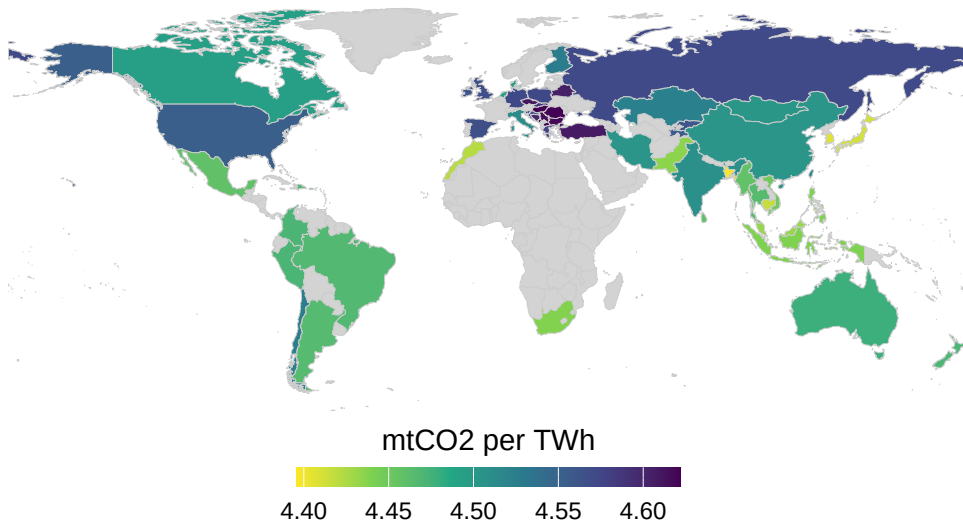
```
ggplot(data = map_data_2024) +
  geom_sf(aes(fill = log_intensity),
    color = "gray80",
    size = 0.1) +
  scale_fill_viridis_c(
    option = "viridis",
    direction = -1,
    name = "mtCO2 per TWh",
    na.value = "lightgray",
    guide = guide_colorbar(
      title.position = "top",
      title.hjust = 0.5,
      barwidth = 10,
      barheight = 0.5,
```

```

    direction = "horizontal")
) +
theme_void() +
theme(
  legend.position = "bottom",
  legend.title = element_text(vjust = 0.5)) +
labs(
  title = "Global Coal Carbon Intensity by Country (2024)" +
  # No Antarctica
  coord_sf(ylim = c(-55, 85))

```

Global Coal Carbon Intensity by Country (2024)



```

ggsave("Global Coal Carbon Intensity by Country (2024).png", width = 6, height=4, units = "in")

```

Modeling Intensity Over Time By Region

```

intensity_time_region_model <- lm(formula = mtCO2_per_TWh ~ Year * Area, data=region_data)
summary(intensity_time_region_model)

```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year * Area, data = region_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0074538	-0.0009099	0.0000789	0.0009910	0.0060173

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.871e+00	1.167e-01	16.026	< 2e-16	***
Year	-5.047e-04	5.802e-05	-8.700	3.71e-15	***
AreaAsia	2.425e+00	1.651e-01	14.689	< 2e-16	***
AreaEurope	-8.000e-01	1.651e-01	-4.846	2.95e-06	***
AreaLatin America and Caribbean	-1.489e+00	1.651e-01	-9.022	5.31e-16	***
AreaMiddle East	-1.022e+00	1.651e-01	-6.189	4.82e-09	***
AreaNorth America	-6.828e-01	1.651e-01	-4.136	5.67e-05	***
AreaOceania	-2.902e-01	1.651e-01	-1.758	0.0807	.
Year:AreaAsia	-1.174e-03	8.205e-05	-14.314	< 2e-16	***
Year:AreaEurope	4.578e-04	8.205e-05	5.580	9.99e-08	***
Year:AreaLatin America and Caribbean	7.552e-04	8.205e-05	9.204	< 2e-16	***
Year:AreaMiddle East	5.095e-04	8.205e-05	6.209	4.35e-09	***
Year:AreaNorth America	3.856e-04	8.205e-05	4.699	5.57e-06	***
Year:AreaOceania	1.581e-04	8.205e-05	1.927	0.0558	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.002092 on 161 degrees of freedom

Multiple R-squared: 0.9978, Adjusted R-squared: 0.9976

F-statistic: 5614 on 13 and 161 DF, p-value: < 2.2e-16

```
shapiro.test(region_data$mtCO2_per_TWh)
```

Shapiro-Wilk normality test

data: region_data\$mtCO2_per_TWh

W = 0.87914, p-value = 1.117e-10

```
old_par <- par(no.readonly = TRUE)
diagnostic_plot_title = "Global Diagnostic Plot"
plot_file_name = paste0(diagnostic_plot_title, ".png")
```

```

png(filename = plot_file_name,
     width = 1500,
     height = 1000,
     units = "px",
     res = 150)
par(mfrow = c(2, 2),
    family = "serif",
    oma = c(0, 0, 2, 0),
    mar = c(5, 5, 2, 1))
plot(intensity_time_region_model, sub=" ")
mtext(text = diagnostic_plot_title,
      outer = TRUE,
      cex = 1.5,
      font = 2,
      line = 0,
      adj = 0)
dev.off()

```

pdf
2

```
par(old_par)
```

```
vif(intensity_time_region_model)
```

there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif

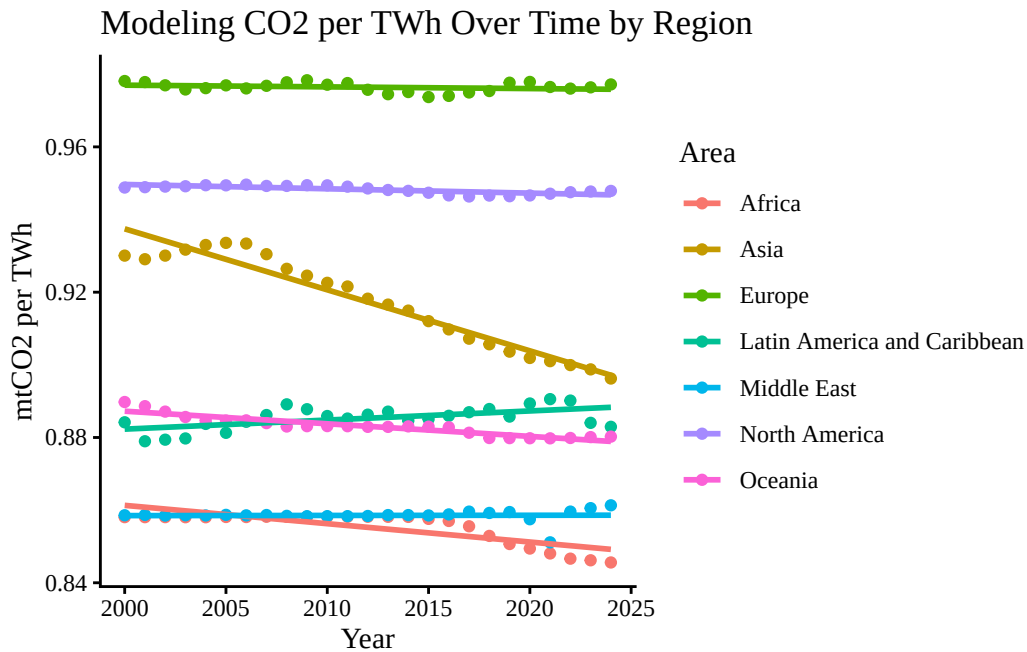
		GVIF	Df	GVIF ^{1/(2*Df)}
Year	7.000000e+00	1		2.645751
Area	2.226123e+29	6		279.015991
Year:Area	2.226294e+29	6		279.017783

```

ggplot(region_data, aes(x=Year, y=mtCO2_per_TWh, color=Area)) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="Modeling CO2 per TWh Over Time by Region",
       y= "mtCO2 per TWh")

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
ggsave("Modeling CO2 per TWh Over Time by Region.png", width = 7, height=4, units = "in", dp
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
region_model_maker_2 <- function(region) {
  df <- region_data %>%
    filter(Area == region)
  old_par <- par(no.readonly = TRUE)
  diagnostic_plot_title = paste(region, "Diagnostic Plot")
  plot_file_name = paste0(diagnostic_plot_title, ".png")

  model <- lm(formula = mtCO2_per_TWh ~ Year, data=df)
  print(summary(model))
  png(filename = plot_file_name,
      width = 1500,
      height = 1000,
      units = "px",
      res = 150)
  par(mfrow = c(2, 2),
```

```

    family = "serif",
    oma = c(0, 0, 2, 0),
    mar = c(5, 5, 2, 1))
plot(model, sub=" ")
mtext(text = diagnostic_plot_title,
      outer = TRUE,
      cex = 1.5,
      font = 2,
      line = 0,
      adj = 0)
dev.off()
par(old_par)
plot(ggplot(df, aes(x=Year, y=mtCO2_per_TWh)) + geom_point())
plot(ggplot(df, aes(x=mtCO2_per_TWh)) + geom_histogram())
}

```

```

regions = c("Asia",
            "Africa",
            "Europe",
            "Latin America and Caribbean",
            "Middle East",
            "Oceania",
            "North America")
for (i in regions){
  df <- region_data %>%
  filter(Area == i)
  print(i)
  print(shapiro.test(df$mtCO2_per_TWh))
}

```

```
[1] "Asia"
```

Shapiro-Wilk normality test

data: df\$mtCO2_per_TWh

W = 0.90794, p-value = 0.02743

```
[1] "Africa"
```

Shapiro-Wilk normality test

```
data: df$mtCO2_per_TWh
W = 0.67129, p-value = 3.066e-06
```

```
[1] "Europe"
```

```
Shapiro-Wilk normality test
```

```
data: df$mtCO2_per_TWh
W = 0.959, p-value = 0.395
```

```
[1] "Latin America and Caribbean"
```

```
Shapiro-Wilk normality test
```

```
data: df$mtCO2_per_TWh
W = 0.95898, p-value = 0.3945
```

```
[1] "Middle East"
```

```
Shapiro-Wilk normality test
```

```
data: df$mtCO2_per_TWh
W = 0.5853, p-value = 2.913e-07
```

```
[1] "Oceania"
```

```
Shapiro-Wilk normality test
```

```
data: df$mtCO2_per_TWh
W = 0.90222, p-value = 0.02051
```

```
[1] "North America"
```

```
Shapiro-Wilk normality test
```

```
data: df$mtCO2_per_TWh
W = 0.88603, p-value = 0.009224
```

```
region_model_maker_2("Asia")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0073744	-0.0015561	-0.0002215	0.0022260	0.0060173

Coefficients:

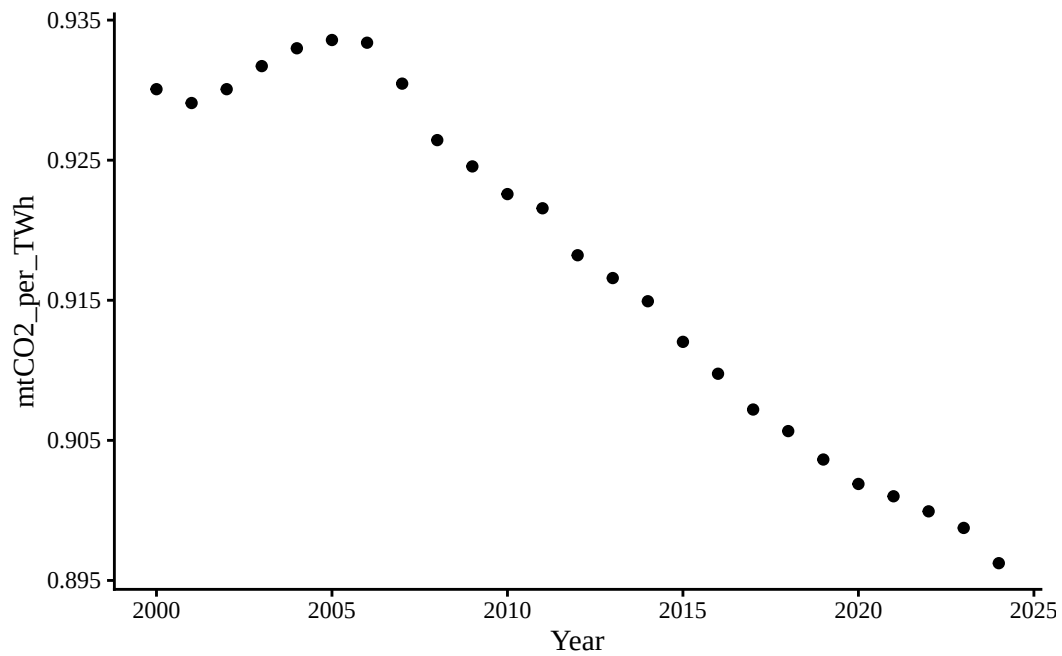
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.296e+00	1.804e-01	23.81	< 2e-16 ***
Year	-1.679e-03	8.968e-05	-18.73	2.02e-15 ***

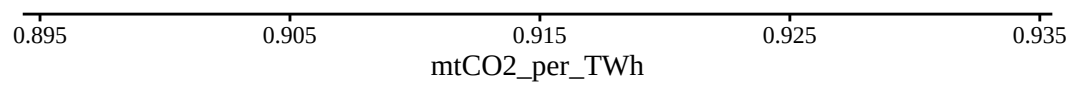
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.003233 on 23 degrees of freedom

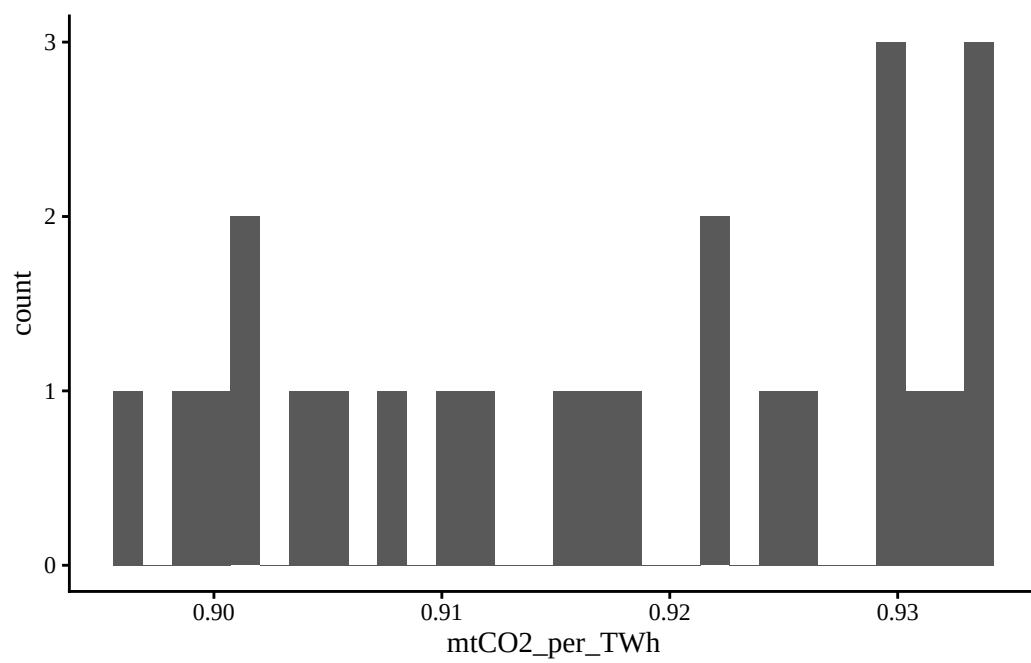
Multiple R-squared: 0.9384, Adjusted R-squared: 0.9358

F-statistic: 350.6 on 1 and 23 DF, p-value: 2.022e-15





``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



```
region_model_maker_2("Africa")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0036120	-0.0022354	-0.0001883	0.0024013	0.0038899

Coefficients:

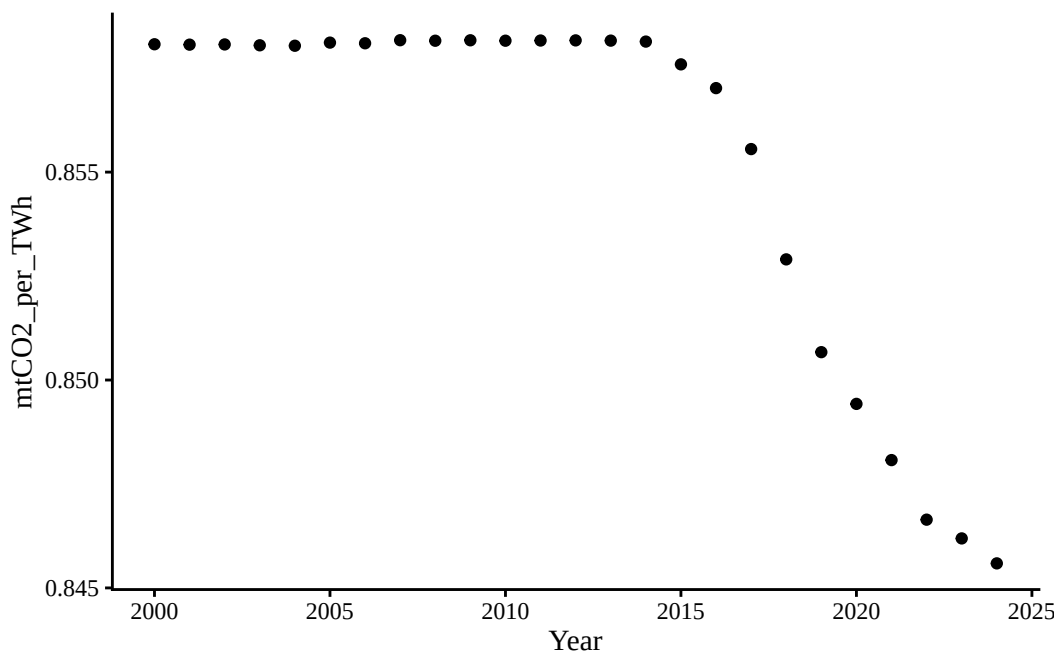
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.871e+00	1.484e-01	12.609	8.17e-12 ***
Year	-5.047e-04	7.374e-05	-6.845	5.58e-07 ***

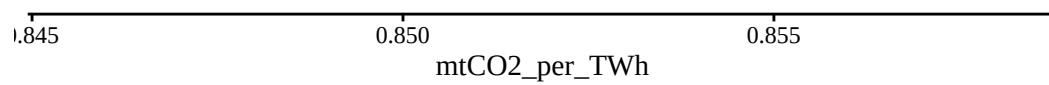
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.002659 on 23 degrees of freedom

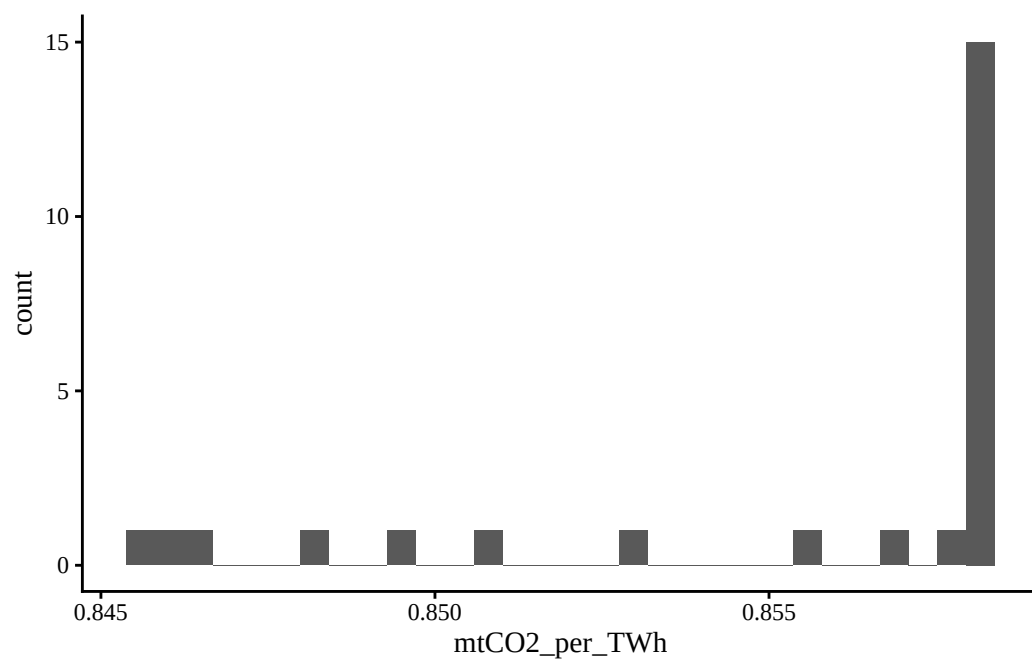
Multiple R-squared: 0.6707, Adjusted R-squared: 0.6564

F-statistic: 46.85 on 1 and 23 DF, p-value: 5.581e-07





``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
region_model_maker_2("Europe")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0025686	-0.0007716	0.0001545	0.0010998	0.0018303

Coefficients:

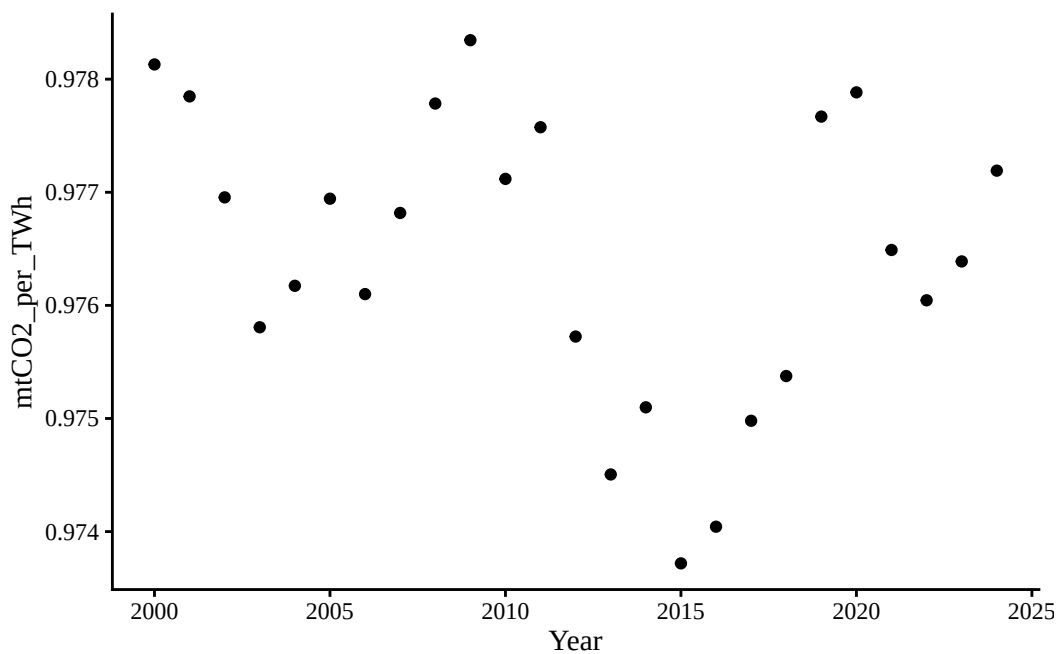
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.071e+00	7.096e-02	15.09	2.02e-13 ***
Year	-4.692e-05	3.527e-05	-1.33	0.197

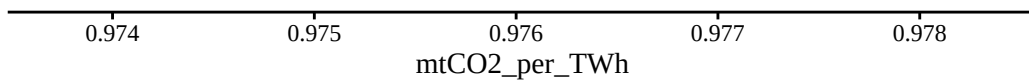
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.001272 on 23 degrees of freedom

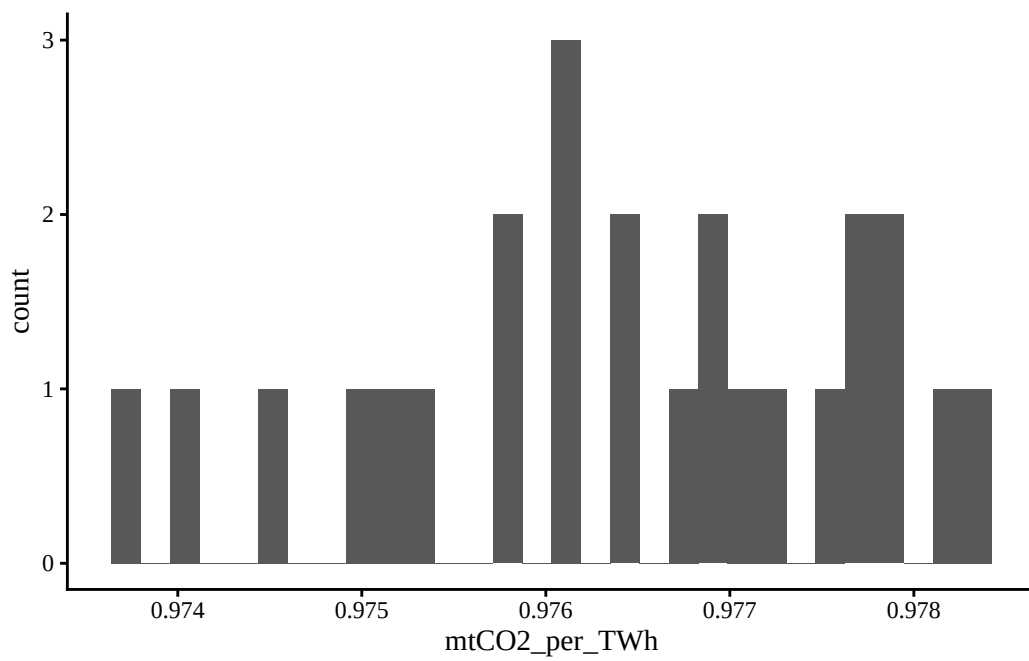
Multiple R-squared: 0.07143, Adjusted R-squared: 0.03106

F-statistic: 1.769 on 1 and 23 DF, p-value: 0.1965





``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
region_model_maker_2("Latin America and Caribbean")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0053731	-0.0012999	0.0004404	0.0018901	0.0048285

Coefficients:

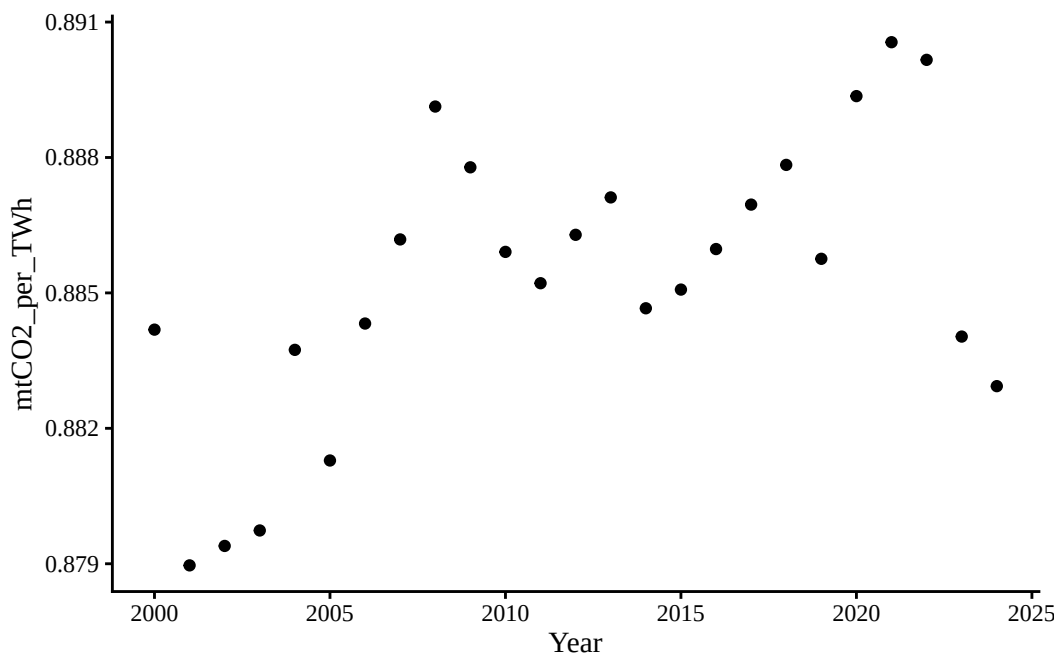
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.814e-01	1.456e-01	2.619	0.01534 *
Year	2.505e-04	7.238e-05	3.460	0.00212 **

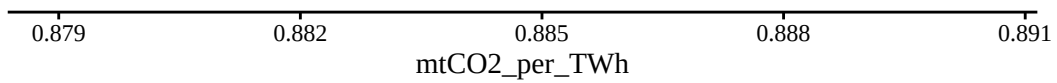
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.00261 on 23 degrees of freedom

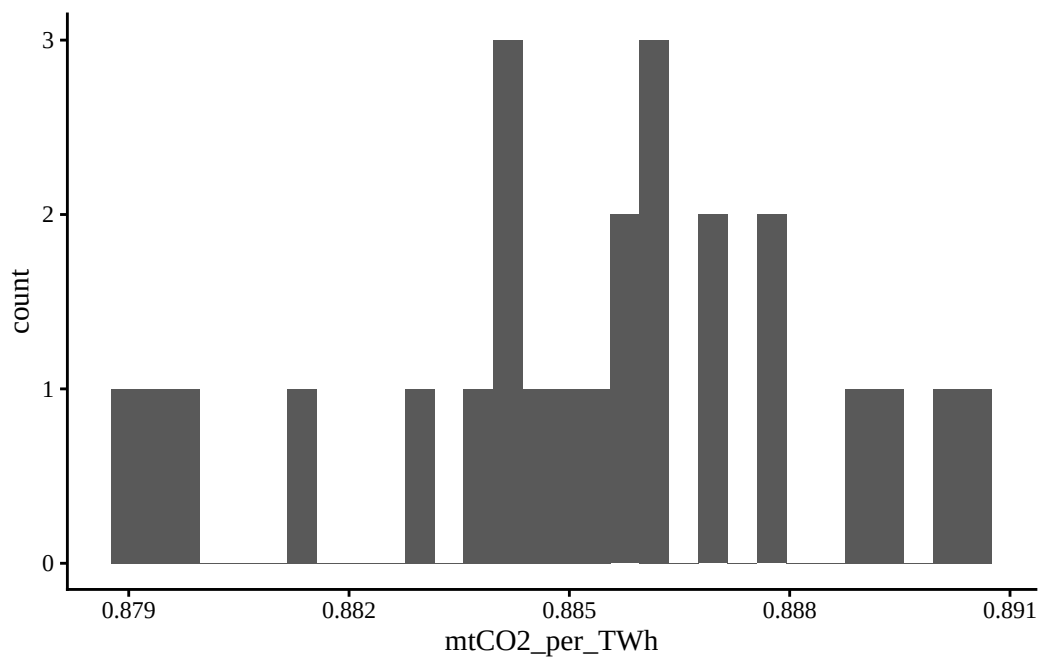
Multiple R-squared: 0.3424, Adjusted R-squared: 0.3138

F-statistic: 11.97 on 1 and 23 DF, p-value: 0.002123





``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



```
region_model_maker_2("Middle East")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0074538	-0.0001060	0.0000839	0.0002576	0.0027242

Coefficients:

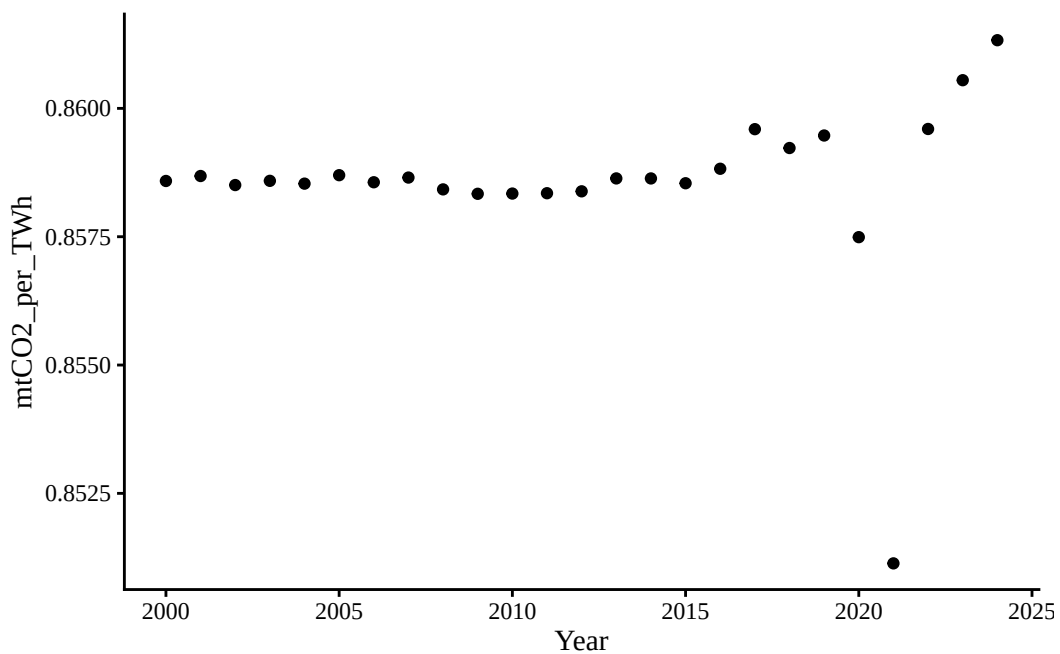
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.490e-01	9.846e-02	8.623	1.16e-08 ***
Year	4.749e-06	4.893e-05	0.097	0.924

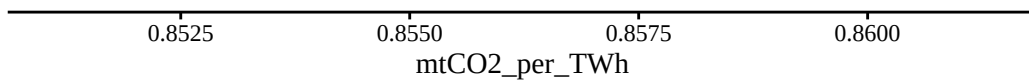
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.001764 on 23 degrees of freedom

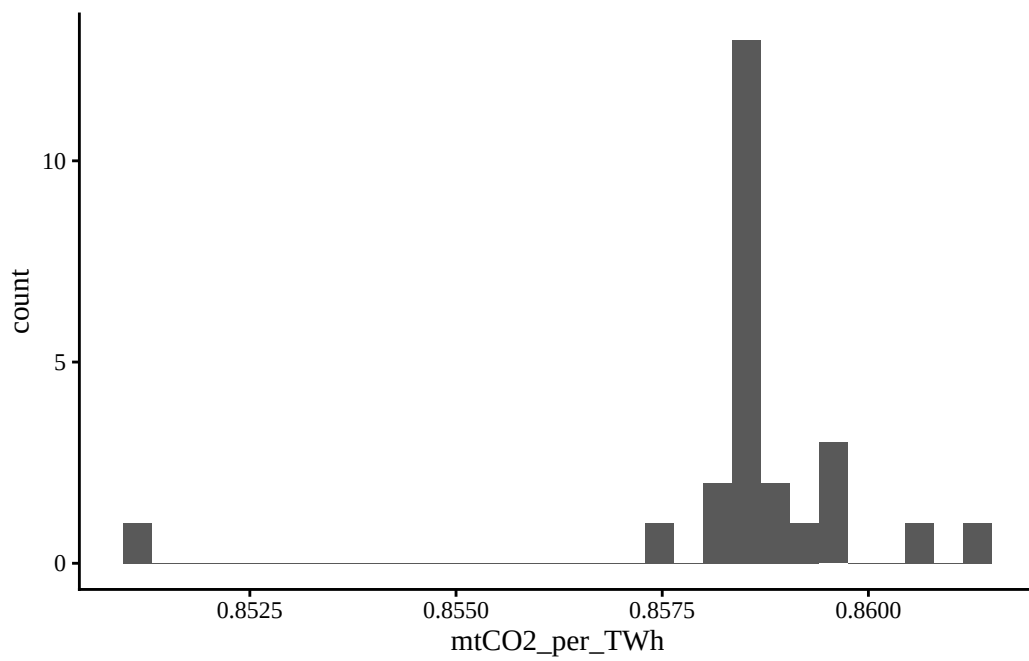
Multiple R-squared: 0.0004093, Adjusted R-squared: -0.04305

F-statistic: 0.009417 on 1 and 23 DF, p-value: 0.9235





``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



```
region_model_maker_2("Oceania")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.0014399	-0.0008311	-0.0001952	0.0006692	0.0025247

Coefficients:

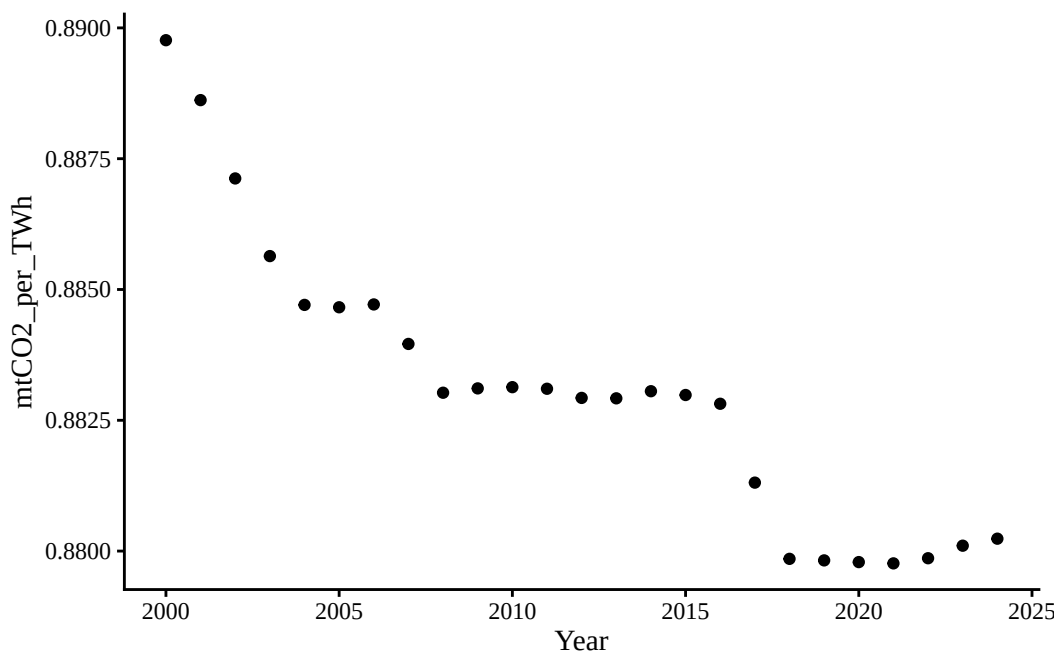
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.581e+00	5.705e-02	27.70	< 2e-16 ***
Year	-3.467e-04	2.836e-05	-12.23	1.52e-11 ***

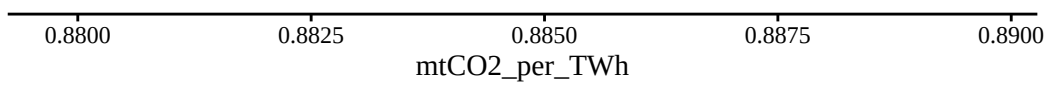
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.001022 on 23 degrees of freedom

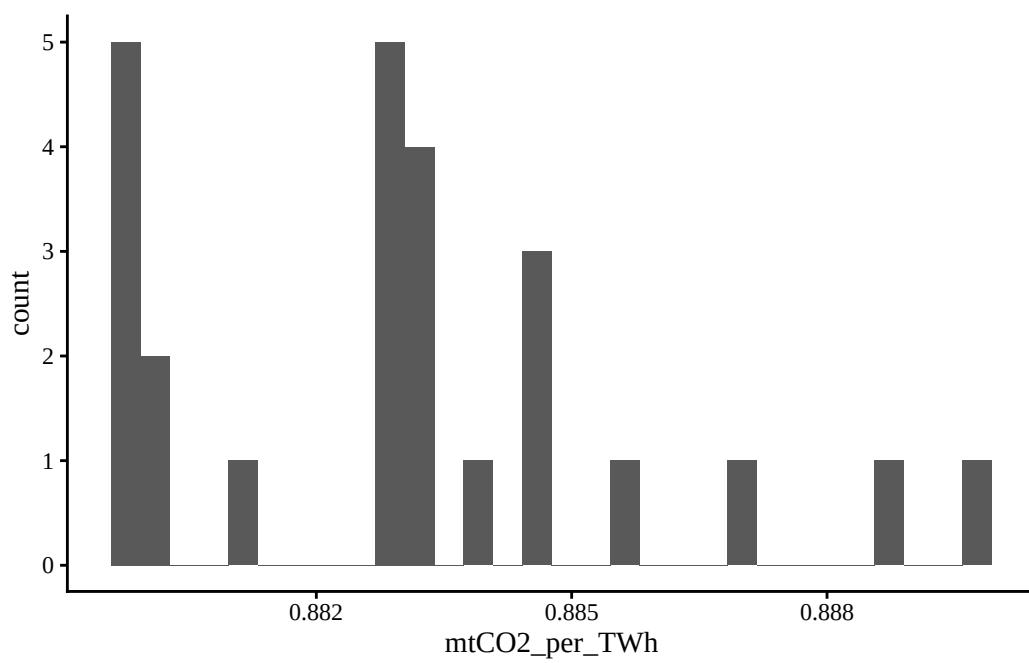
Multiple R-squared: 0.8666, Adjusted R-squared: 0.8608

F-statistic: 149.5 on 1 and 23 DF, p-value: 1.522e-11





``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



```
region_model_maker_2("North America")
```

Call:

```
lm(formula = mtCO2_per_TWh ~ Year, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.288e-03	-6.344e-04	2.737e-05	5.520e-04	1.083e-03

Coefficients:

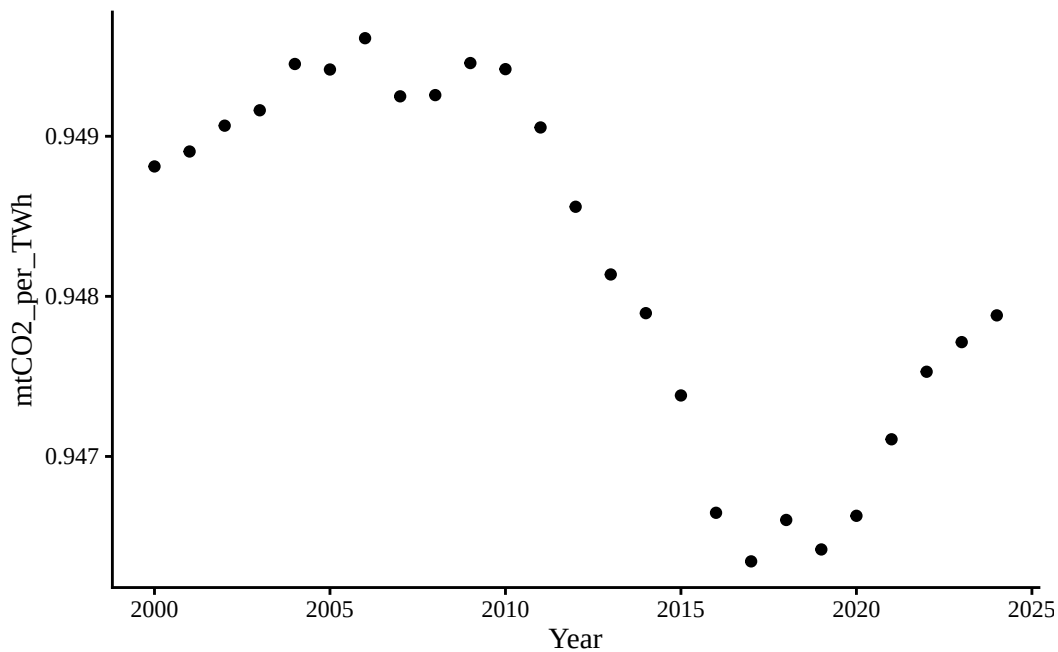
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.188e+00	4.039e-02	29.414	< 2e-16 ***
Year	-1.192e-04	2.007e-05	-5.936	4.73e-06 ***

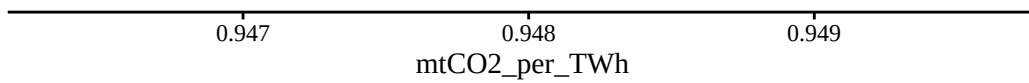
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0007237 on 23 degrees of freedom

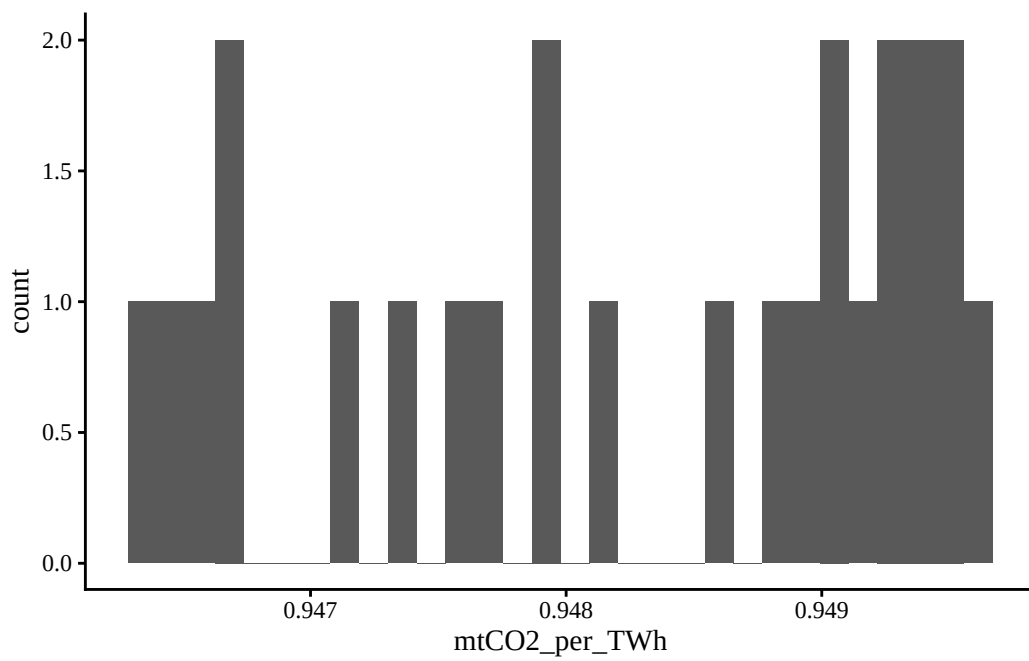
Multiple R-squared: 0.6051, Adjusted R-squared: 0.5879

F-statistic: 35.24 on 1 and 23 DF, p-value: 4.729e-06





``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



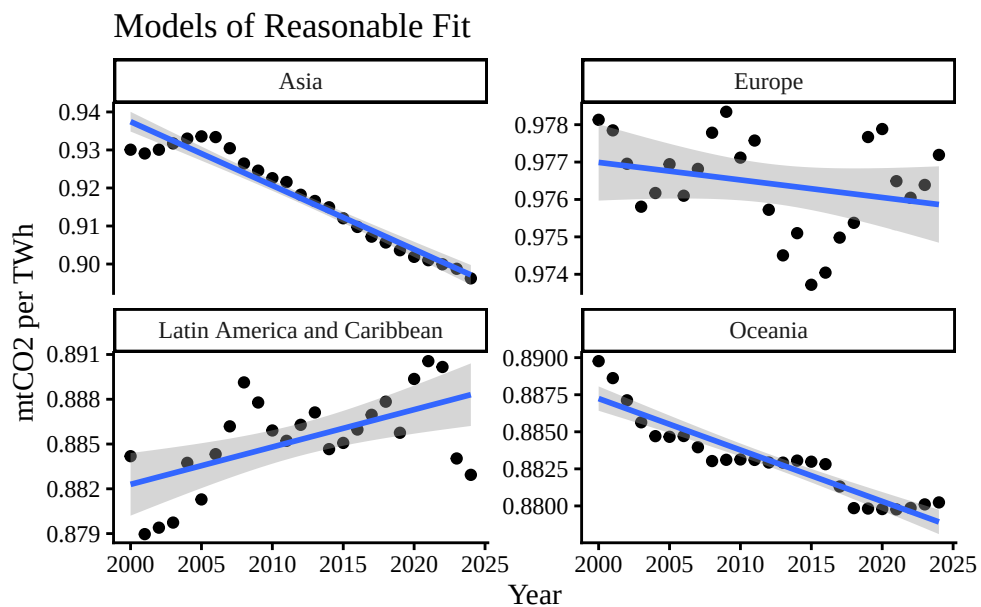
```

good_models_regions <- c("Europe",
                        "Latin America and Caribbean",
                        "Asia",
                        "Oceania")
good_models_data <- region_data %>%
  filter(Area %in% good_models_regions)

ggplot(good_models_data, aes(x=Year, y=mtCO2_per_TWh)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title="Models of Reasonable Fit",
       y = "mtCO2 per TWh") +
  facet_wrap(~Area, scales = "free_y") +
  theme(plot.margin = unit(c(5, 10, 5, 5), "mm"))

```

`geom\_smooth()` using formula = 'y ~ x'



```

ggsave("Models of Reasonable Fit.png", width = 7, height=4, units = "in", dpi = 1500)

```

`geom\_smooth()` using formula = 'y ~ x'